

Telco Capstone Project

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DATA 480

Introduction

Technology has a major impact on how we live and work in the present day. A lot of how we use technology is ingrained in how we communicate, view entertainment, and protect valuable information. My capstone project is about subscription services, and helping businesses figure out the best ways to retain their customers. With so many subscription services competing against each other, this project aims to analyze a businesses' customer base, and determine their habits, dislikes, and likes. By understanding the customer base and their preferences, this newfound knowledge can help other businesses understand which segments have the highest risk of churning. My formal research question is: Which customer segments have the highest churn risk? This question was expanded into three smaller questions. They are: Which subscription plans are most associated with customer churn? What customer behaviors serve as strong predictors of churn?, and Can we build a predictive model for churn?

Data

The data I'm using for my project is a fictional subscription service dataset called Telco. This dataset includes multiple types of subscription services, such as phone, internet, device protection and movie streaming to name a few. The dataset has 7,043 observations with 21 variables. To get the data ready to be analyzed, I removed all of the NA's mentioned in the dataset. To run linear regressions, I converted the non-numeric variables I wanted to analyze into binary values. These included phone service, internet service, tv streaming, movie streaming, online security, online backup, device protection, many lines, tech support and contract length. The variable I wanted to control for was churn, so this was also converted to binary.

Methods

The packages that were used are readr, tidyverse, psych, and dplyr in R. I created tables for all of the variables I wanted to analyze and combined them with the churn variable. For example, the internet service variable was included in a table with the churn variable. This table was called IStab. This helped me keep track of and narrow down which specific variable I want to compare to others. I ran correlations on each table with the psych package. This took the form of $\text{phi}(\text{IStab}, 3)$. Once correlations were run, linear regressions were done to see the significance between tables.

Results

My initial research question was broken into three smaller questions. The first sub-question was: Which subscription plans are most associated with customer churn? To answer this question, I ran linear regression models with each subscription variable and churn to view the significance. This took the form of $m1 \leftarrow \text{lm}(\text{telco}\$Chbin \sim \text{telco}\$PSbin)$, with Chbin representing churn and PSbin representing phone service. Surprisingly, phone service was not significant with churn due to the lack of any stars or small p-value. This might suggest that the phone service plan doesn't have much effect on churn rates. Similar regression models were run with the churn variable and other services, such as internet service, tv streaming, movie streaming, online security, online backup, and device protection.

For internet service, the baseline churn rate for customers with no internet service was 7%. Customers who have the internet are 24 percentage points more likely to churn than those without. It's unlikely that this is rare because of the 3 stars and p-value less than 0.05. TV streaming's base churn rate is 33% for those without the service. Customers with tv streaming are 3 percentage points less likely to churn than those without. Movie streaming's results were the same. Online security's base churn rate was 41% for those without the service. Customers

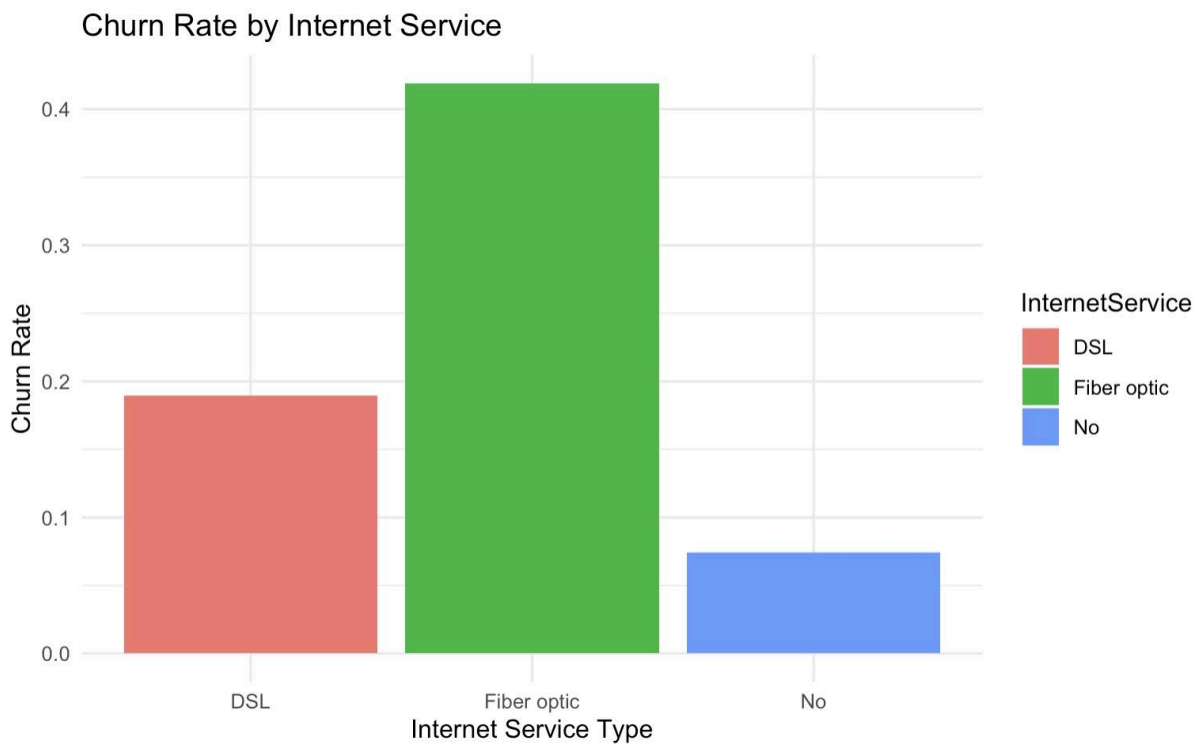
with online security were 27 percentage points less likely to churn than those without. For online backup, customer base churn rate was 39% for those with no online backup. Those with online backup were 18 percentage points less likely to churn. Lastly, for device protection, the base churn rate for customers without device protection was 39%. Customers with device protection are 16 percentage points less likely to churn than those without device protection. All of the linear regressions except for phone service were significant due to small p-values and 3 stars. To summarize, the subscription plans most associated with customer churn are internet service, online backup, device protection, and online security. Internet service has a high positive relationship with churn. Online backup, online security, and device protection have negative relationships with churn. This suggests that many customers prefer keeping their devices protected, while also being dissatisfied with their internet service. To visualize the results of the linear regressions, a table summarizing the findings is below.

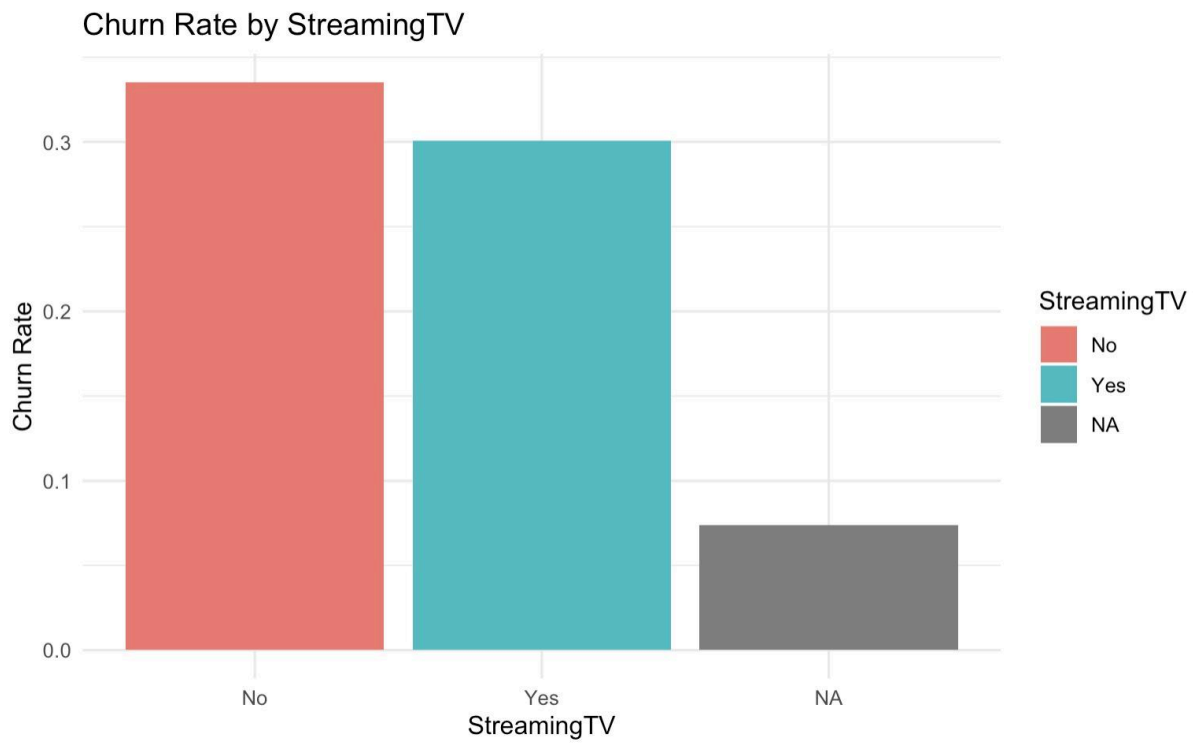
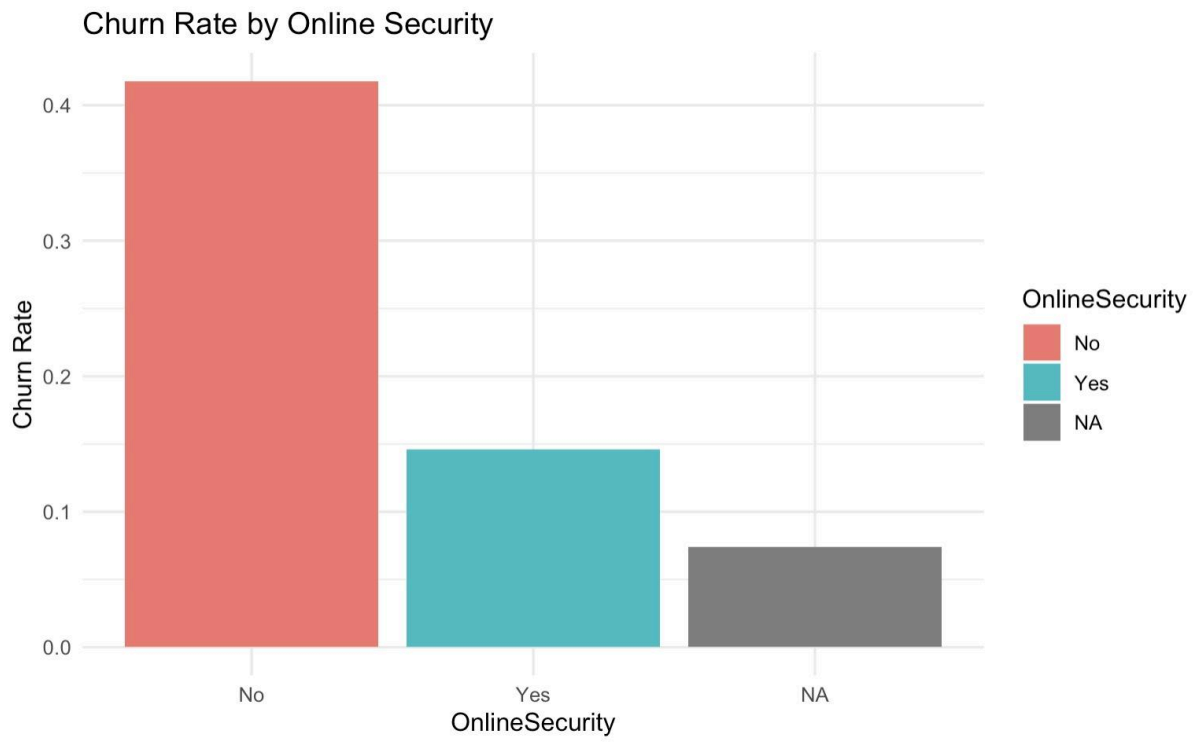
Service	Effect on Churn	Coefficient	p-value	Significance
Internet Service	positive	+0.24	< 2e-16	yes
TV Streaming	negative	-0.03	0.00591	yes
Movie Streaming	negative	-0.03	0.00287	yes
Online Security	negative	-0.27	< 2e-16	yes
Online Backup	negative	-0.18	< 2e-16	yes
Device Protection	negative	-0.16	< 2e-16	yes

Original dataset

My second sub-question was: What customer behaviors serve as strong predictors of churn? After analyzing the above data with linear regressions, we can see that customers that have internet service are statistically more likely to churn compared to those without internet

service. Customers that have protective services, such as online backup, online security, or device protection, are less likely to churn than those that don't have it. So internet service serves as a strong positive predictor of churn, and online security, online backup, and device protection serve as strong negative predictors of churn. Below are some of the churn rates for significant services.



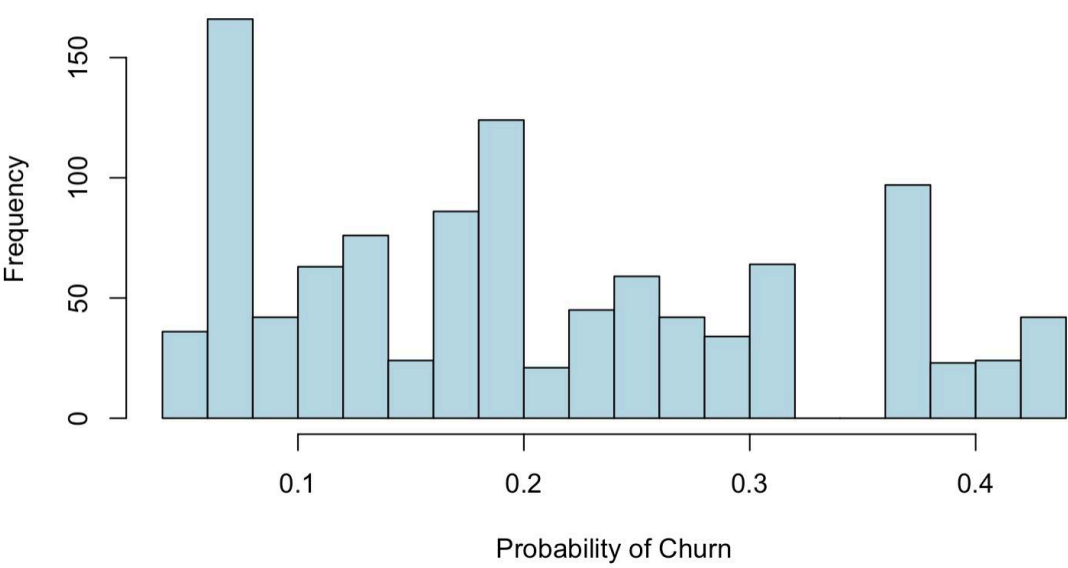


The third sub-question was: Can we build a predictive model for churn? Due to limited time and recent lessons, a logistic regression model was used. All subscription variables were put in the logistic regression. Online security, device protection, and online backup were all significant and negative as expected. Movie streaming and tv streaming were positive. Internet service showed NA's. To try and fix this problem, I dropped all NA's. I ran a new logistic regression with the new clean telco data. These steps predict churn probabilities and determine a cutoff threshold. If the probability is greater than 0.5, then this means "will churn". If the probability is less than or equal to 0.5, then it "won't churn". We can now look at the dataset and see the new columns that are made. Chbin is the original churn, predicted_prob is the model's estimated probability and predicted_class is the model's choice, or if it is churned or not. This answers the question of whether or not a predictive model can be made for churn. After making the predictive model, it was found that the logistic regression model predicted churn rates accurately 79.49% of the time. The table and plots below summarize these results.

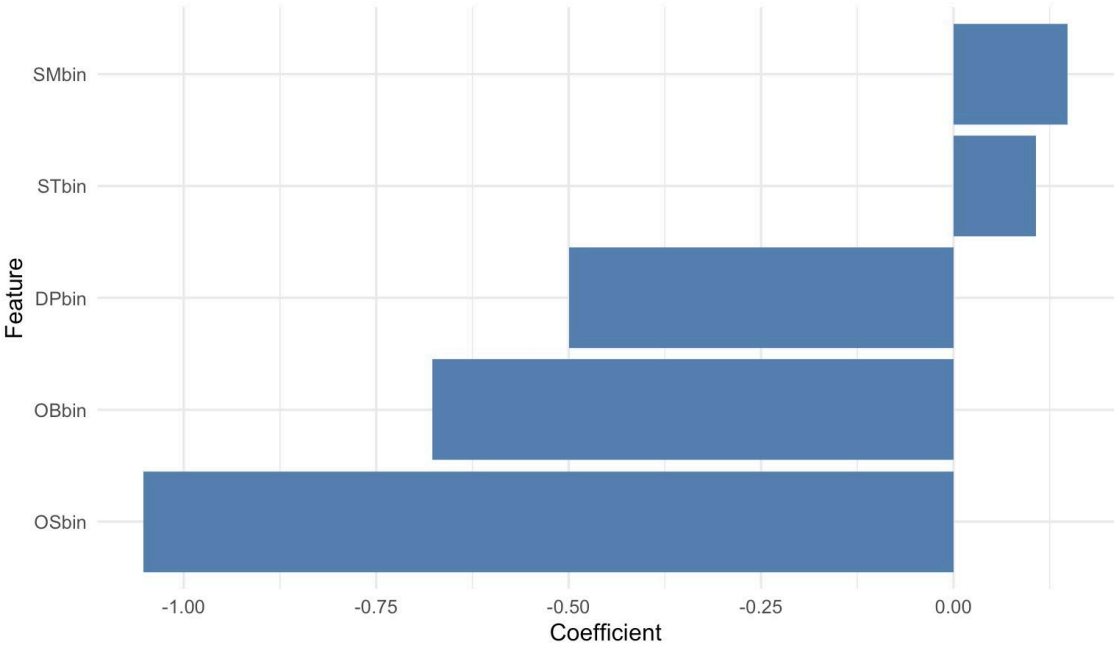
Service	Coefficient	p-value	Effect on Churn
Internet Service	NA	NA	Dropped
TV Streaming	0.10	0.1807	Increases churn
Movie Streaming	0.14	0.1807	Increases churn
Online Security	-1.05	2.23e-09	Decreases churn
Online Backup	-0.67	3.23e-05	Decreases churn
Device Protection	-0.49	0.003	Decreases churn

Cleaned dataset

Predicted Churn Probabilities



Feature Importance (Logistic Coefficients)



Limitations

Some of the limitations of my work are the dataset, time frame, and methods used to evaluate the data. The Telco dataset is fictional, and doesn't include real data from subscription companies. If there was a combined dataset of real companies with phone, internet, movie, and device protection services, many of the results would look different. In addition to this, one dataset doesn't represent the many challenges businesses face when it comes to analyzing their customer base and working on retention strategies for customer churn. To fully understand churn rates as it relates to subscription services, many more datasets could be used to paint a bigger picture.

Another limitation is the time frame. The capstone project has to be doable within four months. If more time was allotted to this project, more complicated predictive models could've been used such as classification or clustering. The methods used to evaluate the data also limit its scope. All analysis was done in R. Using other packages in R or using tools such as SQL, Tableau, or C++ would make the results appear differently.

Reflection

If I were to do the capstone project again, I would choose a smaller dataset to work with. Instead of 7,000 observations, I would choose values closer to 1,000 or 2,000. I would also explore a different topic. Instead of leaning into my business minor, I would do a topic related to college students, such as determining which factors impact getting a job after graduation. I would also challenge myself and use SQL or Python instead of R. R was more convenient since I was currently working with it, but I would use Python if I was able to redo the project. Additionally, I would learn how to use a machine learning model to analyze my data. I didn't get the chance to try this with my current project, but I have an interest in learning more about K-Nearest Neighbors or Decision Trees and how they work.